Banking Multi-Agent Support Bot

Sophomore-level project documentation template

Architecture: User → BiGRU classifier → deterministic router → specialist AgentExecutor → tools/RAG → LLM → Streamlit.

1. Selected BANKING77 intents

Intent	Agent
card_payment_fee_charged	Billing
cash_withdrawal_charge	Billing
card_not_working	Technical
order_physical_card	Booking / card-order
country_support	General

2. Dataset

BANKING77 contains 13,083 online banking customer-service queries and 77 fine-grained intents. This project filters five labels from the training portion and creates a stratified 80/10/10 split.

3. Deep-learning classifier

Trainable word embedding (100 dimensions) → bidirectional GRU (128 hidden units) → dropout → linear classifier. Cross-entropy loss and Adam optimizer. The checkpoint is selected using validation macro-F1.

4. Evaluation results

Test accuracy: _____ Test macro-F1: _____

Insert artifacts/confusion_matrix.png and artifacts/f1_by_intent.png after training.

5. LangChain agents and tools

Billing: check_invoice + search_docs. Technical: search_docs. Booking/card-order: check_availability + search_docs. General: search_docs. Each agent has a separate system prompt. The classifier chooses the specialist; the LLM does not choose the route.

6. RAG

12 local Markdown articles are split into chunks, embedded with sentence-transformers/all-MiniLM-L6-v2, stored in FAISS, and retrieved through LangChain.

7. Memory and reliability

Streamlit session state stores conversation history and passes it as LangChain chat history. The app has a 20-request cap, short prompts, top-3 retrieval, a three-iteration agent limit, Groq→Ollama fallback, and an offline response if both providers fail.

8. AI disclosure

The source files include explicit AI-GENERATED SECTION START/END markers. Preserve these markers and follow the instructor policy.